

The Dynamic Aging Mind: Revelations From Functional Neuroimaging Research

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Abstract

The conception of the aging mind that emerged from behavioral and structural imaging studies portrayed the mind as a victim of passive deterioration and decline with age, with a few domains of preserved function. The advent of functional neuroimaging has demonstrated that the aging brain is an adaptive and plastic structure that responds dynamically to cognitive challenge and structural deterioration—thus, fundamentally changing views of cognitive aging. In addition, a neural theory of the aging mind based on behavioral data—the dedifferentiation view of cognitive aging—was largely confirmed when neuroimaging technology became available to test it. We argue that functional neuroimaging has advanced cognitive aging theories by creating a stronger emphasis on compensatory mechanisms related to brain plasticity and potential reorganization as evidenced by the resurgence of interest and research in cognitive training research designed to improve cognition through enhancement of neural structures or reorganization of functional circuitry.

Keywords

aging, fMRI, compensation, plasticity, scaffolding

In this article, we consider how the advent of functional neuroimaging has either confirmed or changed cognitive theories of the aging mind that had been initially developed from behavioral findings and structural brain imaging. This issue of how functional imaging impacted theories of cognitive aging is particularly interesting because cognitive aging, by its very nature, is a neurobiological event, so it stands to reason that new findings about how the brain functions might change views of how cognitive mechanisms function and differ with age. Inherent in this entire discussion is recognition that even behaviorally derived theories of cognitive aging have often considered underlying brain structures, although relatively little attention has been paid to neural function until the advent of functional imaging.

Theories of cognitive aging generally agree on the behavioral phenomena to be explained. A vast literature shows that as adults age, many domains of cognitive function decline, including speed of processing, working memory, long-term memory, and reasoning (Hertzog, Dixon, Hultsch, & MacDonald, 2003; Park & Reuter-Lorenz, 2009; Salthouse, 1996), all of which decline at similar rates (Park et al., 1996). In addition, some domains of cognitive function are preserved with age, including verbal ability (Park et al., 1996), implicit or procedural memory (Howard, 1988), and prospective memory in naturalistic settings (Henry, MacLeod, Phillips, & Crawford, 2004). Finally, it is well-recognized that as cognitive tasks become more demanding, age differences in performance often become larger (Myerson, Hale, Wagstaff, Poon,

& Smith, 1990). Most behavioral theories of cognitive function address all of these findings.

We will selectively review three major theoretical accounts of cognitive aging that focus on different behavioral mechanisms as the basis for cognitive aging, and have served as the conceptual basis for much research. We also briefly describe behavioral and structural imaging data that have provided support for the theories. The three theories we discuss are (a) speed of processing (Salthouse, 1996), (b) limited cognitive resource (Craik & Byrd, 1982), and (c) dedifferentiation (Lindenberger & Baltes, 1994). Common to all three of these theories is the notion that a fundamental mechanism is driving much of the age-related decline in multiple domains of cognitive function. We also note that it is difficult to consider theories of the aging mind that evolved much later than 1995, as neural mechanisms were readily integrated into views of cognitive aging once neuroimaging data became available.

We argue that the advent of functional neuroimaging fundamentally changed the view of the aging mind from a passive model of decline to a dynamic model of adaptation characterized by plasticity and reorganization of function in response to

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neural degradation and cognitive challenge. A brief summary of each theory appears below.

Cognitive Aging Theories

Limited resource theory

This influential view was advanced by Craik and colleagues (Craik & Byrd, 1982), who suggested that a fixed pool of cognitive resources available for processing information systematically declined with age. Tasks that required fewer resources (e.g., implicit memory) showed a minimal effect of age (if any), whereas tasks that required a greater amount of resources (e.g., free recall) showed large age differences. Age-related declines could be mitigated through various types of environmental supports that relieved processing demands (e.g., supportive encoding cues or supportive retrieval tasks). As resource theory evolved, an independent measure of cognitive resource was conceptualized to be working memory (Park et al., 1996). Hasher and Zacks (1988) proposed a different perspective in which inefficient inhibitory mechanisms limited working memory capacity, resulting in impaired abilities such as maintaining or switching attention to relevant information. Data from neuropsychological testing and structural imaging suggested that resource demanding tasks heavily relied on the frontal cortex, which shows some of the greatest age-related decline in structural volume (Raz et al., 2005)—this is consistent with the notion of a shrinking pool of resources.

Speed of processing

According to speed of processing theory (Salthouse, 1996), age-related declines in cognition resulted from decreased processing speed, which has two components: limited time for operations and simultaneity of operations. The limited-time component refers to older adults' execution of fewer processing operations in the same amount of time as young adults, thus limiting the quality or accuracy of later processing. The simultaneity mechanism refers to older adults' decreased availability of relevant information from early processing steps—due to rapid decay or displacement—that cannot be appropriately applied to later processing steps. The more complex the task, the more processing steps needed, and thus the more these two mechanisms compound to reveal greater age-related differences (the complexity effect; Cerella, Poon, & Williams, 1980). Age effects can be modulated by changes in strategy and reliance on semantic knowledge (Salthouse, 1996). With this theory, considerable thought has been given to underlying neural mechanisms that might explain age-related slowing. These mechanisms included loss of myelination in white matter tracts, delayed neuronal firing, and reduced synchronization of neural activation (Salthouse, 1992). However, structural imaging provides only limited support for these underlying neural mechanisms. For example, there is evidence that processing speed declines as white matter hyperintensities increase (Gunning-Dixon & Raz, 2000; Schmidt et al., 1993) and as white matter structural

connectivity decreases (O'Sullivan et al., 2001; Stebbins et al., 2001). Nevertheless, the amount of mediated variance from these studies is small relative to the magnitude of the effects attributed to speed in the cognitive literature.

Dedifferentiation of cognitive function

In a study using a sample of adults ranging in age from 70 to 103, Lindenberger and Baltes (1994) reported the surprising finding that, as age increased, measures of sensory function (corrected vision and hearing) shared an increasing amount of variance with cognition, attenuating more age-related variance than even speed. The notion that vision or audition could predict cognition was not predicted by either speed of processing or resource theories. Lindenberger and Baltes hypothesized that a common cause mediated declines in both cognitive and sensory mechanisms with age. They proposed that the cause was less differentiation or specificity in neural mechanisms due to age-related brain degradation resulting in the “dedifferentiation” of behavioral and sensory function. Thus, they posited a clear hypothesis about neural function before functional neuroimaging data was commonly available for the study of cognition. It is worth noting that structural imaging revealed little, if any, age-related differences in the volume of primary visual processing regions in the occipital cortex (Raz et al., 2005), but computational modeling of the theory provided additional support for it (Li & Lindenberger, 1999). Nevertheless, the strong test of this theory could only be provided by functional neuroimaging data.

Summary of cognitive aging theories

The dominant view of the aging mind from 1980s and 1990s was a model of declining cognitive behavior and degrading brain structure. The observation that some aspects of cognition were less susceptible to aging was interpreted as being due to a reliance on accrued knowledge or fewer processing requirements as measured by speed (Salthouse, 1996) or working memory (Park et al., 1996; Salthouse & Babcock, 1991). The idea that cognition could be changed or improved in older adults was focused on the notion that one could teach shifts in strategy or provide environmental supports that lowered processing demands (Craik & Byrd, 1982). Compensation was viewed as an active, conscious attempt to utilize alternative strategies to achieve cognitive goals (Bäckman & Dixon, 1992; Baltes & Baltes, 1990). The concept of *reserve capacity*—a pool of resources that could be accessed or drawn upon late in life—was advanced by Baltes and Baltes (1990), but there was sparse empirical evidence to support the notion.

Functional Neuroimaging Results and Theories of Cognitive Aging

The theories described above are focused on cognitive mechanisms of decline and offered little speculation about age

differences in neural activity, although it certainly seems likely that these theories would have predicted decreases in neural activity with age. In 1999, the National Research Council developed a report entitled “The Aging Mind” and identified three research domains most likely to lead to scientific breakthroughs in the study of cognitive aging (Stern & Carstensen, 2000). This prescient report heralded the integration of behavioral and neural mechanisms to understand cognitive aging. Two of the three areas focused on this integration, calling for concerted efforts to understand what constitutes neural health in the aging brain, and also to develop models of the neural structure of the aging mind. We argue that this promise has been, and continues to be, realized, as even the earliest neuroimaging studies showed surprising findings that have changed our conception of the aging mind. We discuss below some major findings from the aging/functional imaging literature below and how these findings affect behavioral views of cognitive aging.

The aging brain is characterized by higher levels of neural activity in prefrontal regions

Older adults often show bilateral prefrontal activations on both working memory and long-term memory tasks whereas younger adults show primarily left-lateralized prefrontal activations (Cabeza et al., 1997; Grady, McIntosh, Rajah, Beig, & Craik, 1999; Reuter-Lorenz et al., 2000). This increased neural activity with age would not have been predicted by any of the behavioral theories discussed and is generally viewed as compensatory recruitment of additional neural resources that maintain cognitive performance (Cabeza et al., 1997, Reuter-Lorenz et al., 2000). This additional recruitment has played a dominant role in recognizing the dynamic nature of the aging mind, although interpretations are still debated. An integrative theory of the aging mind that builds on the notion of compensatory neural activity is the *scaffolding theory of aging and cognition* (Park & Reuter-Lorenz, 2009), which suggests that the aging brain is a plastic, adaptive structure that responds to age-associated degradation of neural structures by reorganizing, increasing, and distributing neural function to maintain cognitive performance with age, a view quite different from the earlier behavioral views.

As task difficulty increases, older adults reach a maximum level of neural activity sooner than young adults

Studies parametrically manipulating cognitive load have revealed that as task difficulty increases, older adults reach a maximum level of neural function sooner than do young adults—often in the prefrontal cortex (PFC; Cappell, Gmeindl, & Reuter-Lorenz, 2010; Nagel et al., 2009;). This finding suggests that neural resources reach a ceiling for activation earlier and at lower levels of difficulty in older adults than in young

adults (compensation-related utilization of neural circuits or CRUNCH; Reuter-Lorenz & Cappell, 2008). The CRUNCH model aligns most closely with limited resource theory but proposes that the brain is responding dynamically to cognitive challenge by selectively activating more neural resources than young adults do to maintain performance. The resource model, in contrast, postulates an ever-shrinking passive pool of resources with age that is increasingly inadequate to maintain cognitive function.

Implicit memory is age invariant, yet there is clear evidence for more neural activity with age

Implicit memory tests have been regarded largely as age invariant by cognitive theories of aging, implying that cognitive abilities engaged by these tasks are protected from effects of aging (Moscovitch & Winocur, 1992). In word-stem completion (Daselaar, Veltman, Rombouts, Raaijmakers, & Jonker, 2005) and repetition priming (Bergerbest et al., 2009) tasks, young adults showed reduced neural activity—interpreted as greater neural efficiency—for primed words in the left inferior frontal cortex (IFC), whereas older adults showed reduced neural activity in both the left and right IFC, which were positively correlated with priming performance. These findings illustrate a large contrast between predictions generated from both speed of processing and resource theories and functional neuroimaging data. The notion that older adults might actually require more neural resources to perform an equivalent task is not obvious from these theories.

Older adults show proportionally more activity in frontal regions but less activity in posterior regions on cognitive tasks than young adults

In addition to the greater neural activity in frontal regions (mentioned above), older adults frequently show reduced activity in posterior regions, including occipitotemporal regions related to sensory processing. For instance, Grady et al. (1994) investigated face and location perception and showed age-related increases in the PFC and decreases in occipitotemporal regions. Subsequent research has extended these findings to episodic memory and named this pattern the *posterior-anterior shift in aging* (PASA; Davis, Dennis, Daselaar, Fleck, & Cabeza, 2008). These findings have been interpreted as compensatory as higher order cognitive processes are recruited to offset deficits in sensory processing. We would argue that the dominant cognitive aging theories did not address the potentially important role that different types of resource pools could play in determining sensory function—that is, the notion that young adults rely more on occipitotemporal-sensory processes whereas older adults rely more on frontal-strategic processes.

Older adults show increases in PFC and medial temporal lobe (MTL) functional connectivity, but decreases in MTL and parietotemporal connectivity in comparison with young adults

A consistent finding that is beginning to emerge in research is age-related differences in functional connectivity or coactivation of brain regions over time. Whereas young adults frequently reveal functional connectivity between the MTL and parietotemporal regions during episodic memory tasks, older adults reveal functional connectivity between the MTL and PFC regions during the same tasks (Daselaar, Fleck, Dobbins, Madden, & Cabeza, 2006; Dennis et al., 2008). Like PASA, these findings have been interpreted as evidence for compensatory reorganization in frontal regions in response to declines in the posterior cortex. These findings are most related to predictions made by the speed of processing theory, which predicted age-related declines in functional connectivity. However, this theory focused on a whole brain decline in connectivity rather than the reorganization in specific regions as demonstrated by neuroimaging.

Specialization of neural function decreases in older adults compared with young adults

Regions of ventral visual cortex that are specialized for face and scene processing become less specialized with advanced age (Grady et al., 1992; Park et al., 2004). Subsequent studies have revealed that these decreases in specialization are associated with decreases in processing speed and other higher order cognitive processes in older adults (Park, Carp, Hebrank, Park, & Polk, 2010). These predictions stem directly from the dedifferentiation theory that made insightful and testable predictions about the aging brain that have been verified through extensive neuroimaging research. The neuroimaging data provide much more detail about underlying mechanisms than was possible from a behavioral theory and has extended the theory to nonsensory regions, including the PFC (Carp, Park, Polk, & Park, 2011).

Conclusion

Advanced age is accompanied by deterioration of cognitive processes. However, brain function does not passively decline with age, as most cognitive aging theories would have predicted, but rather the aging brain retains plasticity and adapts to these challenges by recruiting additional neural resources and/or reorganizing neural networks to best utilize the available neural circuitry (Davis et al., 2008; Park & Reuter-Lorenz, 2009).

We note that broader psychosocial views of aging advanced by Baltes and Baltes (1990) had a much stronger view of the plasticity and dynamism of the aging system compared with other cognitive theories, and that this perspective has ultimately proven to be more consistent with results from neuroimaging

studies than have basic cognitive aging theories. However, even Baltes and Baltes (1990) viewed compensation resulting from age as a conscious change in strategy or a use of external aids, rather than a nonconscious change in the neural substrates responsible for maintaining cognitive processes.

As pointed out in the most recent Handbook of the Psychology of Aging, "Many theories of cognitive aging—even local theories of memory, attention, executive function, emotion regulation, well-being, personality—are incorporating multiple aspects of the neurological substrate in their research programs." (Dixon, 2011, pp. 16). Recent cognitive theories of aging have neural compensation as a basic component, invoking mechanisms such as cognitive reserve (Stern, 2009), functional plasticity (Greenwood, 2007), and neural scaffolding (Park & Reuter-Lorenz, 2009). Such strong views of neural plasticity have resulted in a resurgence of interest in cognitive training research designed to improve cognition through enhancement of neural structures or reorganization of functional circuitry (Lustig, Shah, Seidler, & Reuter-Lorenz, 2009). Functional imaging of the aging mind has provided a fundamental shift in contemporary views of aging, shifting from a picture of static decline to a more dynamic model with considerable complexity and reorganization underlying the inevitable changes in cognition that do occur with age.

Declaration of Conflicting Interests

The authors declared that they had no conflicts of interest with respect to their authorship or the publication of this article.

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